GATE-DA

Subject – (Artificial Intelligence)

Mock Test – 1 (Answer Key)

Topics Covered

Search: Informed, Uninformed

Adversarial; Logic, Propositional, Predicate;

Reasoning under uncertainty topics — conditional independence representation

Exact inference through variable elimination

Approximate inference through sampling.

Section A – Artificial Intelligence

Q	Answer
1	**B**
2	**A, C, D**
3	**B**
4	**1024**
5	**A, C**
6	**C**
7	**1093**
8	**A, B, C**
9	**B**
10	**B**
11	**B**

Q Answer

12 **256**

13 **B, C, D**

14 **B**

15 **C**

16 **B**

17 **A, C, D**

18 **256**

19 **B**

20 **C**

21 **A, B, C, D**

22 **B**

23 **64**

24 **C**

25 **B**

26 **A, B, C**

27 **16**

28 **B**

29 **A, B, C**

30 **A**

SECTION B – Aptitude

Q	Answer
31	**A**
32	**106**
33	**C**
34	**B**
35	**C**
36	**A, B, D**
37	**7.20**
38	**B**
39	**A** (equivalently 10/66 = 20/132 = 5/33)
40	**B**

Quick Verification of Numerical Questions

Q4

Leaf nodes = $4^5 = 1024$

Q7

Total nodes:

$$1 + 3 + 9 + 27 + 81 + 243 + 729 = 1093$$

Q12

Binary tree leaves:

$$2^8 = 256$$

Q18

Truth table rows:

$$2^8 = 256$$

Q23

Interpretations:

$$2^6 = 64$$

Q27

Binary variable with 3 binary parents:

$$2^3 \times 2 = 16$$

(CPT entries)

Q32

$$x^2 + y^2 = (x + y)^2 - 2xy$$
$$= 14^2 - 2(45) = 196 - 90 = 106$$

Q34

Distance:

$$180 + 220 = 400$$
$$v = 400/20 = 20 \text{ m/s}$$

Q37

$$\frac{1}{T} = \frac{1}{12} + \frac{1}{18} = \frac{5}{36}$$
$$T = \frac{36}{5} = 7.2$$

Q38

Total sum:

$$8 \times 15 = 120$$

Remaining:

$$120 - 27 = 93$$

Average:

$$93/7 = 13.29$$

Q39

$$\frac{5}{12} \times \frac{4}{11} = \frac{20}{132} = \frac{10}{66} = \frac{5}{33}$$

Two corrections worth noting

1. **Q39** has **three equivalent options** (A, B, and C are the same probability). That makes the question flawed. The correct probability is:

$$\boxed{\frac{5}{33}}$$

2. **Q40** should ideally be re-enumerated carefully because arrangement-count questions can be sensitive to interpretation.